

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Equation of a Line

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3, 3)$, $B(0, -2)$, $C(-4, -2)$ and $D(-1, 3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC . **ICSE 2025, 5 marks**

SOLUTION

(a) The diagonals of a parallelogram bisect each other, so P is the mid-point of AC : $P = \left(\frac{3 + (-4)}{2}, \frac{3 + (-2)}{2} \right) = \left(-\frac{1}{2}, \frac{1}{2} \right)$.

(b) Slope of $CB = \frac{-2 - (-2)}{0 - (-4)} = 0$ and slope of $DA = \frac{3 - 3}{-1 - (-4)} = 0$. The slopes are equal, so $CB \parallel DA$.

(c) Slope of $AC = \frac{3 - (-2)}{3 - (-4)} = \frac{5}{7}$. Using $A(3, 3)$: $y - 3 = \frac{5}{7}(x - 3)$, so $7y - 21 = 5x - 15$.

Answer: (a) $P\left(-\frac{1}{2}, \frac{1}{2}\right)$ (b) both slopes are 0 (c) $5x - 7y + 6 = 0$

2. Triangle ABC has vertices $A(1, -1)$, $B(0, 4)$ and $C(-6, 4)$, and D is the mid-point of BC . Find (a) the coordinates of D (b) the equation of the median AD . **ICSE 2023, 3 marks**

SOLUTION

(a) $D = \left(\frac{0 + (-6)}{2}, \frac{4 + 4}{2} \right) = (-3, 4)$

(b) Slope of $AD = \frac{4 - (-1)}{-3 - 1} = -\frac{5}{4}$. Through A : $y + 1 = -\frac{5}{4}(x - 1)$, so $4y + 4 = -5x + 5$.

Answer: (a) $D(-3, 4)$ (b) $5x + 4y - 1 = 0$

3. $A(2, 5)$, $B(-1, 2)$ and $C(5, 8)$ are the vertices of a triangle. M is the point on AB with $AM : MB = 1 : 2$. Find M , and then the equation of the line through C and M . **ICSE 2018, 4 marks**

SOLUTION

$M = \left(\frac{1(-1) + 2(2)}{3}, \frac{1(2) + 2(5)}{3} \right) = (1, 4)$

Slope of $CM = \frac{8 - 4}{5 - 1} = 1$, so $y - 4 = 1(x - 1)$.

Answer: $M(1, 4)$; the line is $y = x + 3$

2 Lines parallel or perpendicular to a given line

4. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ **ICSE 2025, 1 mark**

SOLUTION

A parallel line has the same x and y coefficients: $3x + 4y + k = 0$.

It passes through $(0, 0)$, so $k = 0$.

Answer: (d)

5. The line PQ meets the positive x -axis at P , 5 units from the origin, and the negative y -axis at Q , 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ . Find (a) the coordinates of P and Q (b) the equation of RS . **ICSE 2025, 3 marks**

SOLUTION

(a) $P(5, 0)$ and $Q(0, -3)$.

(b) Slope of $PQ = \frac{0 - (-3)}{5 - 0} = \frac{3}{5}$, so the slope of RS is $-\frac{5}{3}$.

RS passes through the origin: $y = -\frac{5}{3}x$.

Answer: (a) $P(5, 0)$, $Q(0, -3)$ (b) $5x + 3y = 0$

6. (a) The lines $kx - y + 4 = 0$ and $2y = 6x + 7$ are perpendicular. Find k . (b) Find the equation of the line parallel to $2y = 6x + 7$ that passes through $(-1, 1)$. **ICSE 2024, 3 marks**

SOLUTION

(a) The first line is $y = kx + 4$, with slope k . The second is $y = 3x + \frac{7}{2}$, with slope 3.

Perpendicular lines: $k \times 3 = -1$, so $k = -\frac{1}{3}$.

(b) A parallel line has slope 3: $y - 1 = 3(x + 1)$

Answer: (a) $k = -\frac{1}{3}$ (b) $y = 3x + 4$

7. The lines $5x - 3y + 2 = 0$ and $6x - py + 7 = 0$ are perpendicular. Find p . Then find the equation of the line through $(-2, -1)$ parallel to $6x - py + 7 = 0$. **ICSE 2020, 3 marks**

SOLUTION

Slopes: $\frac{5}{3}$ and $\frac{6}{p}$. Perpendicular: $\frac{5}{3} \times \frac{6}{p} = -1$, so $p = -10$.

The second line is $6x + 10y + 7 = 0$, with slope $-\frac{3}{5}$.

Parallel line through $(-2, -1)$: $y + 1 = -\frac{3}{5}(x + 2)$, so $5y + 5 = -3x - 6$.

Answer: $p = -10$; the line is $3x + 5y + 11 = 0$

8. The vertices of $\triangle ABC$ are $A(3, 8)$, $B(-1, 2)$ and $C(6, -6)$. Find (a) the slope of BC (b) the equation of the line through A perpendicular to BC . **ICSE 2019, 3 marks**

SOLUTION

$$(a) \text{ Slope of } BC = \frac{-6 - 2}{6 - (-1)} = -\frac{8}{7}$$

$$(b) \text{ The perpendicular line has slope } \frac{7}{8}: y - 8 = \frac{7}{8}(x - 3), \text{ so } 8y - 64 = 7x - 21.$$

$$\text{Answer: (a) } -\frac{8}{7} \text{ (b) } 7x - 8y + 43 = 0$$

9. The lines $3x - 5y = 7$ and $4x + ay + 9 = 0$ are perpendicular. Find a . **ICSE 2018, 3 marks**

SOLUTION

$$\text{Slopes: } \frac{3}{5} \text{ and } -\frac{4}{a}. \text{ Perpendicular: } \frac{3}{5} \times \left(-\frac{4}{a}\right) = -1$$

$$12 = 5a$$

$$\text{Answer: } a = \frac{12}{5}$$

10. $A(-1, 3)$, $B(4, 2)$ and $C(3, -2)$ are the vertices of a triangle. Find (a) the centroid G (b) the equation of the line through G parallel to AC . **ICSE 2017, 3 marks**

SOLUTION

$$(a) G = \left(\frac{-1 + 4 + 3}{3}, \frac{3 + 2 - 2}{3} \right) = (2, 1)$$

$$(b) \text{ Slope of } AC = \frac{-2 - 3}{3 - (-1)} = -\frac{5}{4}. \text{ The line is } y - 1 = -\frac{5}{4}(x - 2), \text{ so } 4y - 4 = -5x + 10.$$

$$\text{Answer: (a) } G(2, 1) \text{ (b) } 5x + 4y - 14 = 0$$

3 Inclination and slope

11. The points $A(x, y)$, $B(3, -2)$ and $C(4, -5)$ are collinear. Write y in terms of x . (a) $3x - 11$ (b) $11 - 3x$ (c) $3x - 7$ (d) $7 - 3x$ **ICSE 2024, 1 mark**

SOLUTION

$$\text{Slope of } BC = \frac{-5 + 2}{4 - 3} = -3. \text{ Collinear points give the same slope for } AB.$$

$$\frac{y + 2}{x - 3} = -3, \text{ so } y = -3x + 9 - 2 = 7 - 3x$$

$$\text{Answer: (d)}$$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.